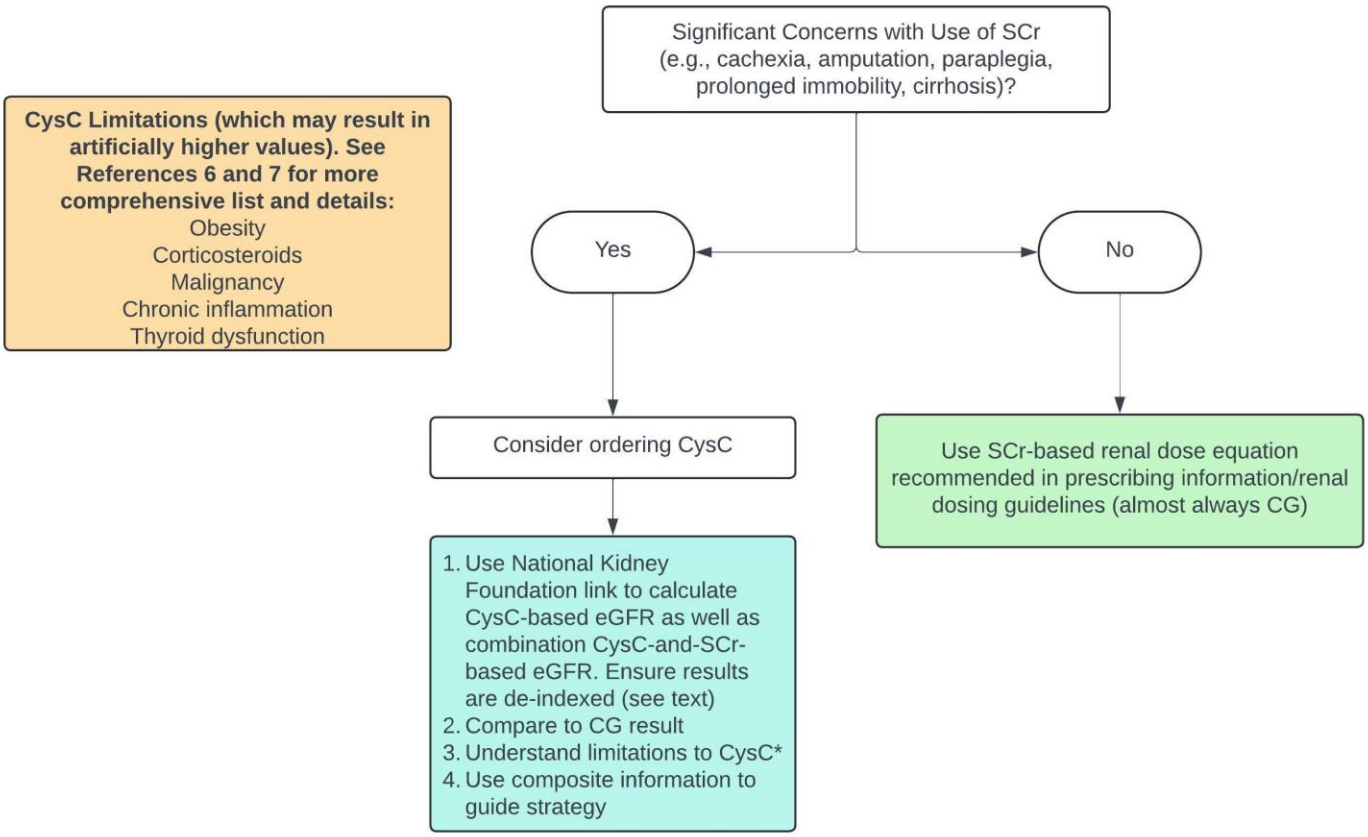
MICHIGAN MEDICINE
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CONSIDERATIONS IN THE USE OF RENAL DOSING EQUATIONS IN ADULTS



Summary:

- We do not recommend universal or default use of CysC to inform renal dosing of medication. We further do not recommend that CysC-based eGFR be utilized in isolation in medication dosing. Rather, we recommend that, in patient populations where SCr-based equations have 1) major limitations, AND 2) a change in estimate of renal function would result in a therapeutic change (i.e., medication dosing, avoidance of medications, avoidance of procedures/contracst, etc) CysC adjunctive assessment may be **considered**. There is generally no role for routine serial measurements (i.e. daily) or measurements if there are not therapeutic implications due to potential discrepancy of CysC and SCr.

- If CG and CKD-EPI values results in discordant dosing recommendations, clinical judgement should be used to inform the most optimal dose and treatment decisions. This decision should be based on the patient's individual factors, including trajectory of renal function, clinical status, medication-specific considerations such as likelihood of toxicity, and severity of illness.

- GFR/eGFR equations are derived from patients with stable renal function. These equations are not validated, nor are they reliable, in the setting of acute kidney injury, fluctuating renal function, or immediately post-kidney transplant before a new steady-state is achieved. Decision-making for dose selection based on a renal estimation equation should take into consideration baseline values and urine output, as well as trends in these parameters.

Additional details:

- The use of renal function equations in medication dosing information evolves over time. As a result, some drugs may have renal dose adjustments that were developed using alternative or retired equations. For antibiotics, such variances will be noted in our renal dosing guidelines. For non-antibiotics, please refer to tertiary references and/or prescribing information.

- Cockroft Gault (CG) is the most widely validated equation for medication dosing. When utilizing CG, we recommend using the following 'rules'. Note that these are pre-built into the CrCl that shows on the left-side of a patient's chart in MiChart:
    - Actual body weight should be used for non-overweight patients
    - For overweight patients (defined as BMI >/= 25), use adjusted body weight (using 0.4 fudge factor).

- The 2021 CKD-EPI equation has been adopted by nephrologists for kidney disease staging but has not been sufficiently validated to comprehensively replace CG for drug dosing. There are three 2021 CKD-EPI equations: 1) SCr-based, 2) CysC-based, 3) combination SCr/CysC-based. The eGFR displayed in laboratory results alongside SCr is calculated via the 2021 CKD-EPI equation.

- Cystatin C (CysC) is a protein released from all nucleated cells and filtered at the glomerulus with no tubular secretion.
    - In general, incorporation of CysC into GFR estimations is likely to result in more accurate estimates of GFR compared to SCr in patients with reduced muscle mass (as a result of cachexia, amputation, advanced age, low body weight, prolonged critical illness, or other factors).
    - However, pharmacists should be aware of several limitations to the use of CysC. *In general, while the limitations to SCr tend to result in overestimation or higher values of GFR (for example, in cachexia), the limitations of Cystatin C tend to result in underestimation or lower values of eGFR.* For example, cancer (especially multiple myeloma) may result in variable and increased CysC release. Obese patients may have elevated CysC levels as adipose tissue expresses more CysC than other cells. Corticosteroids and inflammation have been shown to result in elevated Cystatin C values. Note that these limitations are not comprehensive, and reference lists should be consulted for further information.

    - **As such, at this time, we do not recommend comprehensive use of CysC to inform renal dosing of medications. In addition, we do not recommend that CysC-based eGFR be utilized in isolation in medication dosing. Rather, we recommend that, especially in patient populations where SCr-based equations have 1) major limitations, AND 2) a change in estimate of renal function would result in a therapeutic change (i.e., medication dosing, avoidance of medications, avoidance of procedures/contracst, etc) CysC adjunctive assessment may be considered. There is generally no role for routine serial measurements (i.e. daily) or measurements if there are not therapeutic changes to be made by a potential discrepancy of CysC and SCr.** CysC serum test is available to order in MiChart and is populated in 'Results' with a concordant eGFR value (from the 2012 CKD-EPI equation). This value is standardized to 1.73m^2, and as a result requires standardization/adjustment to compare to CG values. This may be performed at this National Kidney Foundation link: https://www.kidney.org/professionals/gfr_calculator. This link can also provide values for the 3 CKD-EPI 2021 equations referenced above. **If CG and CKD-EPI values results in discordant dosing recommendations, clinical judgement should be utilized to inform the most optimal dose and treatment decisions. This decision should be based on the patient's individual factors, including trajectory of renal function, clinical status, medication-specific considerations, and severity of illness**.

- Other considerations:
    - **GFR/eGFR equations are derived from patients with stable renal function. These equations are not validated, nor are they reliable, in the setting of acute kidney injury, fluctuating renal function, or immediately post-kidney transplant before a new steady-state is achieved. Decision-making for dose**

selection based on a renal estimation equation should take into consideration baseline values and urine output, as well as trends in these parameters.

- Consider adjusting SCr upward in patients with **decompensated cirrhosis** (defined as presence of ascites, esophageal varices, and/or hepatic encephalopathy)[2]

- For **pregnant** patients, a SCr ≥ 0.8 in a pregnant patient should prompt concerns for renal insufficiency, and urine output should be assessed (normal ~ 2-2.5 mL/kg/hr). This is because SCr is generally lower in pregnant patients than non-pregnant patients of the same age given increased blood volume and increased perfusion to the kidneys.

- For **immediate post kidney transplant recipients (within ~7 days),** Scr trend and UOP is more important than the actual Scr. Per clinical judgement, appropriate to dose with est CrCl for ~50-60 ml/min (unless estimated renal function higher based on actual Scr) immediately post-op (e.g., POD 0-2), if Scr is down trending and good UOP (>0.5 ml/kg/h). If Scr is not down trending or has stabilized, dose as appropriate based on estimated renal function per SCr or HD plan. Do not renally dose adjust standard prophylaxis agents (ie valganciclovir, acyclovir, and Bactrim) without consulting with transplant pharmacist.

- Dosing in **Critically Ill Adults**: Renal function is dynamic in the critically ill population. SCr lags behind actual GFR by at least 24- 72 hrs (t ½ SCr is 4 hrs in normals, 8 hrs in CKD Stage II, and 2-3 days CKD Stage IV). Thus, it is important to assess not only serum creatinine and associated renal function equations, but also other clinical factors as appropriate (i.e. urine output/volume status, hemodynamic stability/perfusion, etc.) when determining renal function in the critically ill, as well as trends in those parameters. Under-dosing of antimicrobials in the critically ill has potential negative implications, as critically ill patients often present with septic shock and timely, optimally-dosed antimicrobial therapy near the onset of sepsis/septic shock is likely most impactful. When treating critically ill patients, especially those in shock, it is advisable to dose more aggressively upfront to fill the volume of distribution and optimize PK/PD parameters if renal function is on the threshold of dose adjustment in the absence of trends suggesting worsening function. Vancomycin and aminoglycoside therapeutic drug monitoring may also be helpful in the assessment of renal function. Aminoglycosides, especially, are ideal exogenous filtration markers because they are eliminated entirely by glomerular filtration without tubular reabsorption or secretion. As such, aminoglycoside clearance is an accurate representation of GFR (at the time that the levels were drawn).
    - Within the critically ill patient population, changes in the need for renal replacement therapy (RRT) specifically either the initiation of CRRT in patients with reduced GFR or switching from CRRT to iHD or watchful waiting, often requires significantly different medication dosing regimens. Extra effort should be paid to these patients to ensure medication regimens are optimized during the initiation and discontinuation of RRT therapies. Anticipation of the initiation or discontinuation of CRRT should lead to proactive discussions with the interdisciplinary team to discuss important changes in medications.

- Whether dosing an **antibiotic** by the any equation, the dose recommended based on estimated renal function should often serve as a starting point to development of an optimal recommendation for a specific patient. Clinical status with consideration of pharmacokinetic and pharmacodynamics parameters (MIC of infecting pathogen, severity/site/inoculum of infection) should be factored into decision-making when renal estimation equations provide eGFR or CrCL values that straddle differing dose recommendations. Clinical judgement using a multifaceted approach incorporating an evaluation of trends in renal function and clinical status should supersede negligible variances in calculated renal estimation values.

Last revised: 01/2025	Antimicrobial Stewardship Subcommittee: N/A	P&T: N/A

References

	Citation
1	Lew AK, et al. Evolution of Equations for Estimating Renal Function and Their Application to the Dosing of New Antimicrobials. Ann Pharmacother 2020;54:496-503.
2	Scappaticci GB, Regal RE. Cockcroft-Gault revisited: new de-liver-ance on recommendations for use in cirrhosis. World J Hepatol. 2017;9:131-138.
3	Winter MA, Guhr KN, Berg GM. Impact of various body weights and serum creatinine concentrations on the bias and accuracy of the Cockcroft-Gault equation. Pharmacotherapy. 2012;32:604-612.
4	Wilhelm SM, Kale-Pradhan PB. Estimating creatinine clearance: a meta-analysis. Pharmacotherapy. 2011;31:658-664.
5	Park EJ, Wu K, Mi Z, et al. A systematic comparison of Cockcroft-Gault and modification of diet in renal disease equations for classification of kidney dysfunction and dosage adjustment. Ann Pharmacother. 2012;46:1174-1187.
6	Teaford HR, Barreto JN, Vollmer KJ, et al. Cystatin C: A Primer for Pharmacists. Pharmacy 2020;8:35.
7	St Peter WL, Bzowyckyj AS, Anderson-Haag T, et al. Moving forward from Cockcroft-Gault creatinine clearance to race-free estimated glomerular filtration rate to improve medication-related decision-making in adults across healthcare settings: A consensus of the National Kidney Foundation Workgroup for Implementation of Race-Free eGFR-Based Medication-Related Decisions. Am J Health-Syst Pharm 2025 (in press). https://doi.org/10.1093/ajhp/zxae317

Revision History

Date	Revision
1/2025	Created New Template

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source